10/15 Multiple Seq. Alignment in AlphaFold 2

Why Mult. seq. Align (MSA)?

- previously, we've only been looking at 2 at a time
  - ↳ Good, but ✗ comparison gives limited info
  - ↳ Hard to tell if differences are biology related or just comes from random Mutation
    - L> be more confident in finding patterns

Conceptual Example:

- HW 1 question: going from 2 DP to 3 DP table exponentially increases complexity / runtime

- DP table w/ k strings of length n
  - L>

- until # of cells
  - $2^k - 1$ operations per cell
  - Runtime is $(2n)^{1k}$
- TLDR: naive approach isn't realistic

Heuristics for MSA:

- many different ones to improve on

AlphaFold 2:

- What is it?
  - proteins are chains of Amino Acids, order is called the primary structure
  - DL model that predicts 3D structure based on primary structure
- What forces are involved in protein folding?
  - Hydrogen bonding, +/- side chains

Hierarchy

Historical Approaches

- X-Ray Crystallography → look at defraction patters
- Over time w/ protein engineering to find stabilizing residues
- NMR / Mass Spectroscopy
- Lattice models → algos to predict optimal tiling
- Molecular Dynamics (MD)
  ↳ computationally heavy
- Monte Carlo Methods

Alpha Fold 2 uses MSA to align input protein sequence w/other sequences

↳ find conserved regions of amino acids

Co-evolution: if 2 amino acids sit close to each other in folded 3D protein, a mutation in one often forces a matching mutation in other to maintain structural stability

How is MSA done in Alpha Fold 2

↳ profile HMM → trained to output sequences from a protein family with training data

- Then, given a random sequence, what by the chance that this model gets generated?

Profile HMM (Hidden Markov Model)

- different states (Match, Insertion, Deletion), who's the prob. of the next move?
- sequence generating machine

Putting it together:

- AlphaFold 2 creates the genetic database of similar sequences that it generated to train the model (Match, Insert, Delete). Then take query sequence to see most likely state to produce it; from there, its appended onto MSA.